INCH-POUND

MIL-PRF-39012/114B
12 June 2018
SUPERSEDING
MIL-C-39012/114A
3 October 1986

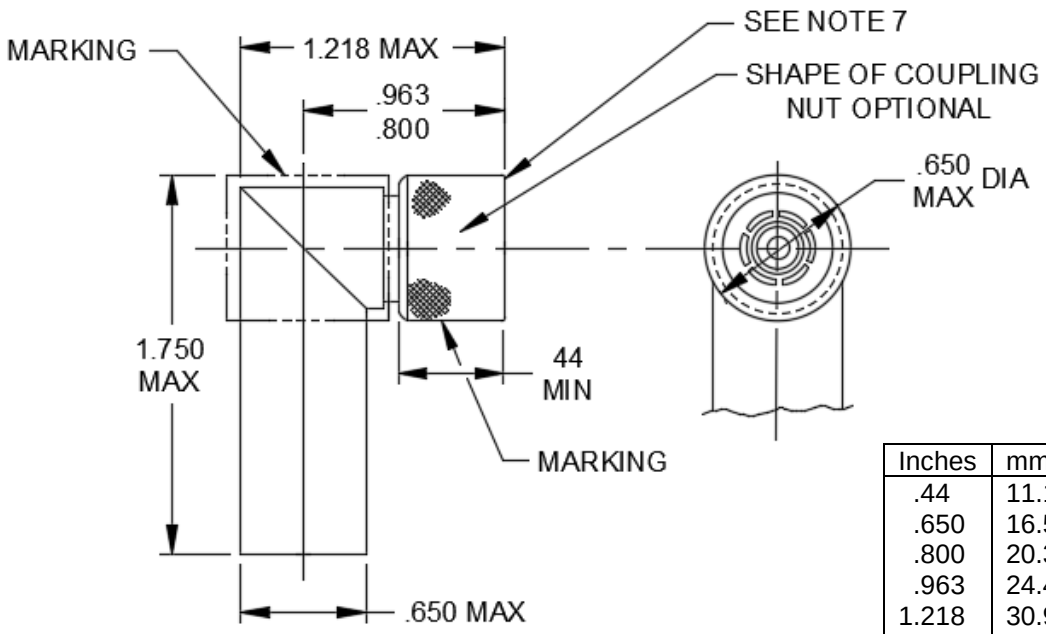
PERFORMANCE SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, COAXIAL, RADIO FREQUENCY (SERIES TNC (CABLED) – PIN CONTACT, RIGHT ANGLE, CLASS 2) FOR SEMI-RIGID CABLE

Reinstated after 12 June 2018 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-PRF-39012.



Inches	mm
.44	11.18
.650	16.51
.800	20.32
.963	24.46
1.218	30.94
1.750	44.45

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension .650 (16.51 mm) is the largest overall diameter of the connector.
4. If applicable, wrench flats are to accommodate standard wrench openings in accordance with FED-STD-H28, appendix 10.
5. Dimension 1.750 (44.45 mm) defines the maximum length of the connector when assembled to the appropriate cable.
6. All undimensioned pictorial configurations are for reference purposes only.
7. Series TNC, pin contact interface per MIL-STD-348.

FIGURE 1. General configuration.

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TABLE I. Dash number and applicable cable.

Dash No	Applicable Cable No.
CATEGORY A – Field Serviceable (no special tools required)	
0001 1/ 2/	M17/130 – RG402
0002 1/ 3/	M17/129 – RG401

1/ These connectors have captivated-center-contacts.
2/ This connector accommodates all cables listed in MIL-DTL-17/130.
3/ This connector accommodates all cables listed in MIL-DTL-17/129.

ENGINEERING DATA.

Nominal impedance: 50 ohms.

Frequency range: 0 to 15,000 MHz.

Voltage rating: 500 volts rms, maximum working voltage at sea level. 125 volts rms, maximum at 70,000 feet.

Temperature rating: -55° to +105°C.

REQUIREMENTS:

Dimensions and construction: See figure 1.

Force to engage and disengage:

Longitudinal force – Not applicable.
Torque – 2 inch-pounds, maximum.

Coupling proof torque: 15 inch-pounds, minimum.

Inspection conditions.

Coupling torque – 4 to 6 inch-pounds.

Mating characteristics: See MIL-STD-348.

Outer contact:

Test ring 10 - .319 maximum, 15 microinch finish.

Insertion depth – 5 pounds, maximum when inserted a minimum of .093.

Contacts with slotted members. Shall contact a .324 minimum diameter ring within .031 of their tip ends.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: MIL-STD-202-302, test condition B. 5,000 megohms, minimum.

Center contact retention: 6 pounds, minimum axial force. Applicable to captivated-center-contact connectors only.

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Corrosion (salt spray): MIL-STD-202-101, test condition B.

Voltage standing wave ratio (VSWR): 1.40 maximum from 2,000 to 15,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower.

Swept frequency VSWR test setup:

Item 6 – VSWR shall be less than 1.02 + .003 F (F in GHz).

Item 16 – VSWR shall be less than 1.02 + .003 F (F in GHz).

Second step of VSWR checkout procedure – VSWR shall be less than 1.06 + .007 F (F in GHz).

Group B inspection – VSWR shall be less than 1.05 + .005 F (F in GHz).

Qualification and group C inspection – VSWR shall not exceed 1.10.

Connector durability: 500 cycles at 12 cycles per minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance. In milliohms maximum.

	Initial	After environment
Center contact	1.5	2.0
Outer contact	0.2	Not applicable
Outer cable conductor to body	0.5	Not applicable

Dielectric withstanding voltage: of MIL-STD-202-301. 1,500 volts rms, minimum at sea level.

Vibration, high frequency: MIL-STD-202-204, test condition B, except the method of mounting shall be approved by the qualifying activity.

Shock: MIL-STD-202-213, test condition I.

Thermal shock: MIL-STD-202-107, test condition B, except test high temperature shall be 115°C.

Moisture resistance: MIL-STD-202-106. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage – 375 volts, minimum.
Altitude – 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency: 1,000 volts rms at a frequency between 5 and 7.5 MHz, inclusive.

Leakage current: Not applicable.

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Cable retention force:

Cable	Pounds (min.)	Torque inch-pounds 1/
M17/130 – RG402	60	3.5
M17/129 – RG401	90	5

1/ Torque shall be applied one cable diameter from the end of the connector. Bend not applicable.

Coupling mechanism retention force: 100 pounds, minimum.

RF leakage: This requirement is met by qualifying the M39012/112 connector for the same cable.

Insertion loss:

.07 $\sqrt{F\ (GHz)}$ dB, maximum tested at 3 GHz.

Part number: M39012/114 (and dash number from table I or "B" number from table II).

TABLE II. Category B – nonfield replaceable (special tools may be required).

M39012/114B 1/ 2/	Applicable Cable No.
0005	M17/130 – RG402
0006	M17/129 – RG401

1/ These connectors have captivated-center-contacts.
2/ For field maintenance purposes these connectors shall be replaced by category A connector.

TABLE III. Cross-reference of part numbers.

Part number 1/ M39012/114B	Superseded part number M39012/114-
0005	0005, 0003
0006	0006, 0004

1/ The new "B" part numbers will be required marking 6 months after the date of this specification. The previous part number may be used in the interim.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-PRF-39012, this document references the following:

FED-STD-H28
MIL-STD-202-101
MIL-STD-202-106
MIL-STD-202-107
MIL-STD-202-204
MIL-STD-202-213
MIL-STD-202-301
MIL-STD-202-302
MIL-STD-348

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force – 85
NASA – NA
DLA – CC

Review activities:
Army – AR, AT, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5935-2018-071)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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